
PREDICTION SETS UNDER HETEROGENEOUS ANNOTATION NOISE: EXACT EFFICIENCY LIMITS FOR BINARY TEXT STRATA

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ABSTRACT

Uncertain annotation channels create ambiguity about a classifier’s target even when noisy-label probabilities are known exactly. We study randomized prediction sets for binary text labels when a fixed stratum map is the only input used at prediction time. For two equally weighted strata with noisy probabilities $(1 - t, t)$, where $0 < t < 1/2$, we determine the minimum expected cardinality at every target marginal coverage $c \in [0, 1]$ under channel rates bounded by t and between-stratum differences bounded by $h \in [0, t]$. Below $h = 1 - 2t$, the frontier has two linear pieces determined by $s = 1 - t/(1 - h)$: its value is c/s for $c \leq s$ and $1 + (c - s)/(1 - s)$ for $c \geq s$. At and above that threshold it equals $2c$, because an observationally compatible fair-label distribution removes the robust efficiency advantage of strata. The optima allow empty sets and are attained by explicit randomized rules. Matching lower bounds apply to every uniformly marginally valid procedure using only the noisy experiment, even when its observable probabilities are known. Synthetic optimization experiments reconstruct this frontier and test its channel sensitivity. The characterization identifies a persistent cost of annotation uncertainty that additional noisy labels alone cannot remove.

1 INTRODUCTION

Prediction sets express uncertainty by returning several plausible labels instead of a single decision. In text classification, the reference label used to calibrate such a set may itself come from an imperfect annotation process. A small set can cover those annotations while failing to cover the intended clean label. This distinction matters for binary tasks such as intent membership or document relevance, even when a trained score is subsequently reduced to a few fixed strata. Work on text uncertainty calibration and conformal NLP motivates explicit coverage objectives (Guo et al., 2021; Campos et al., 2024); noisy-label conformal methods demonstrate that the annotation mechanism enters the coverage analysis (Sesia et al., 2023; Einbinder et al., 2022).

The question studied in this paper concerns the *efficiency cost of uncertainty about that mechanism*. Suppose noisy positive-label probabilities are known in each stratum, and the false-positive and false-negative annotation rates are bounded. Requiring the same rates in all strata couples the possible clean-label distributions. Allowing each stratum to choose its own rates enlarges that family. A set valid for the larger family may need to include additional labels even with unlimited calibration data. Quantifying this persistent cost requires an optimization over compatible clean distributions and an accompanying lower bound over randomized prediction procedures.

We analyze this question in a finite binary model. The text-to-stratum map is fixed independently of calibration, target stratum masses are known, and a prediction rule observes a text only through its stratum. Empty sets are permitted, which makes expected cardinality a linear objective over label-inclusion probabilities. This representation connects the analysis to least-ambiguous set classification and credal prediction (Sadinle et al., 2016;

Javanmardi et al., 2025). The mathematical object of interest is a joint ambiguity family: channel choices in different strata are linked by explicit difference constraints. Replacing that family with separate posterior intervals can destroy usable information about which adverse distributions can occur simultaneously.

Our main result determines the full coverage-size frontier in a symmetric two-stratum experiment. The observable probabilities are $(1 - t, t)$, all four channel rates lie in $[0, t]$, and the difference between corresponding rates is at most h . For $0 < t < 1/2$ and $0 \leq h \leq t$, we identify optimal randomized sets for every coverage level $c \in [0, 1]$. An informative regime has two linear pieces; when h reaches $1 - 2t$, an admissible channel makes the clean label fair in both strata. In that world, coverage equals one half of expected size for every rule. Thus the optimal cost becomes $2c$, exactly the cost of stratum-independent label inclusion. The threshold is feasible only when $t \geq 1/3$.

Two specializations clarify the role of channel structure. At $c = 1 - t$ and $t \leq 1/3$, a common channel permits expected size one, independent stratum channels require size $1 + t$, and a difference bound h yields the exact intermediate value $1 + h$. These are matching attainable costs and observational lower bounds. They concern uniform marginal validity over compatible channels and remain valid when the observable probabilities are supplied exactly.

A companion analysis treats a common channel over any finite number of strata. In an observable overlap regime, the worst coverage of a fixed inclusion policy occurs at one of four channel corners. The population problem is therefore a finite linear program. With stratified noisy calibration samples, a policy-sensitive confidence penalty gives clean-label coverage at least $1 - \alpha$ with calibration probability at least $1 - \delta$, and a cardinality bound approaching the population oracle. The unconditional coverage guarantee is $(1 - \delta)(1 - \alpha)$.

The nearest technical boundary is explicit. Optimizing expected set size through randomized inclusion and finitely many credal constraints is established in Javanmardi et al. (2025); clean posterior oracles are studied by Sadinle et al. (2016), and uncertain-channel calibration by Sesia et al. (2023). Our analysis develops the exact cost as a function of *between-stratum channel tolerance*, including a matching fair-label construction. The numerical study reconstructs that frontier from channel constraints and evaluates the conservatism of finite-sample calibration. Its observations are synthetic; the text interpretation depends on independently justified strata and annotation bounds.

2 RELATED WORK

Expected cardinality under a coverage constraint is a classical efficiency criterion for set-valued classification. Sadinle et al. (2016) derive clean-posterior least-ambiguity rules, including the decomposition of size into label-inclusion probabilities. Adaptive conformal classification and regularized prediction sets combine coverage with more informative set sizes (Romano et al., 2020; Angelopoulos et al., 2020). These methods motivate the objective, but a known clean posterior and a family of observationally equivalent clean posteriors lead to different optimization problems. In the latter, one rule must satisfy several coverage constraints simultaneously.

Randomization does not itself distinguish the present formulation from existing robust prediction. Javanmardi et al. (2025) optimize Bernoulli label inclusion through a vertex-constrained linear program for a credal set, and establish the direction in which enlarging that set changes efficiency. The shared-channel linear program below uses this general principle. Its four constraints arise by reducing a common-denominator fractional channel model, rather than by assuming a supplied list of posterior vertices. The two-stratum frontier then computes the *magnitude* of the efficiency loss as a coupling constraint is relaxed. Generic monotonicity under larger uncertainty sets alone does not give the value $1 + h$ or the threshold $1 - 2t$.

Partial identification provides another formulation of prediction without a unique target law. For interval outcomes, Liu et al. (2025) relate worst compatible coverage to interval containment and optimize set volume under that constraint. Their observation model is a

latent scalar contained in an observed interval, whereas ours observes corrupted binary labels and couples the compatible posteriors across strata. Both formulations optimize against an identified family rather than selecting a single latent distribution. The scientific comparison is consequently about the geometry of that family and the resulting exact optimization, not the general possibility of robust prediction.

Noisy-label conformal prediction already addresses clean-label coverage under annotation contamination. Einbinder et al. (2022) analyze label-noise robustness, while Sesia et al. (2023) develop adaptive classification with noisy labels, including calibration based on uncertainty about a contamination matrix. Their confidence-region construction uses assumptions on inverse-matrix entries and a monotone prediction family. Noise-robust scores and many-class methods further develop this setting (Penso & Goldberger, 2024; Penso et al., 2025). In particular, the uniform-noise concentration analysis in Penso et al. (2025) supplies high-probability clean coverage. The finite-sample result below belongs to this existing confidence-correction approach; its specific ingredients are the shared forward-channel rectangle, a penalty depending on the difference between label-inclusion probabilities, and an explicit expected-size comparison with its population oracle.

Risk-controlling prediction sets and conformal risk control treat losses beyond a single miscoverage event (Bates et al., 2021; Angelopoulos et al., 2022). Constraints on false positives give another criterion for selecting prediction sets (Fisch et al., 2022). These distinctions affect the feasible family: bounding marginal clean-label miscoverage does not imply control of the number of incorrect labels in a set. Conditional conformal guarantees also depend on the collection of conditioning events under consideration (Gibbs et al., 2023). Our guarantees average over a fixed target stratum distribution; conditioning in the calibration result refers to the realized calibration sample, rather than coverage separately at every text or stratum. General conformal background is provided by Angelopoulos & Bates (2021).

Annotation uncertainty also connects to identifiability. Blanchard et al. (2013) study asymmetric label noise through consistency and maximal denoising, and Liu et al. (2022) examine conditions for identifying a label-noise transition matrix. Confident learning estimates uncertainty in dataset labels (Northcutt et al., 2019). The present model instead retains every channel in a stated feasible family. Its lower bounds use the equality of the noisy observation laws, so additional noisy examples alone cannot select between the adverse clean worlds. This does not preclude identification after adding assumptions or independent clean-label information.

For NLP, conformal prediction spans multiple tasks and output types (Campos et al., 2024). Applications include improving zero-shot classification efficiency (Choubey et al., 2022), intent classification with clarification (den Hengst et al., 2024), and uncertainty quantification in multilabel risk-sensitive systems (Hwang et al., 2023). Together with calibration and distillation studies (Guo et al., 2021), these motivate explicit distinctions among scores, annotations and target labels. Their task descriptions provide application context, rather than evidence that binary strata satisfy a common or bounded-heterogeneous annotation channel. The analysis below isolates that structural assumption and computes its consequences within a fully specified binary experiment.

3 TEXT STRATA AND ANNOTATION CHANNELS

Let $Y \in \{0, 1\}$ be the clean text label and $Z \in \{0, 1\}$ its noisy annotation. A fixed measurable map assigns each text to $G \in \{1, \dots, m\}$. The map is chosen independently of the calibration sample, and the known target masses satisfy $\pi_g > 0$ and $\sum_g \pi_g = 1$. Predictions use text only through G . Write

$$p_g = \Pr(Y = 1 \mid G = g), \quad q_g = \Pr(Z = 1 \mid G = g), \quad q_g = a_g + (1 - a_g - b_g)p_g, \quad (1)$$

where $a_g = \Pr(Z = 1 \mid Y = 0, G = g)$ and $b_g = \Pr(Z = 0 \mid Y = 1, G = g)$ are forward channel rates.

A randomized set $S(g) \subseteq \{0, 1\}$ uses fresh randomness independent of the test label conditional on its stratum. Let $r_{gy} = \Pr(y \in S(g))$. Dependence between the two inclusion

events is unrestricted. Every vector $r \in [0, 1]^{2m}$ is attainable by independently including each label with its prescribed probability. In particular, the empty set is an allowed output. Expected cardinality and clean-label coverage are

$$C(r) = \sum_g \pi_g(r_{g0} + r_{g1}), \quad F_p(r) = \sum_g \pi_g\{(1 - p_g)r_{g0} + p_g r_{g1}\}. \quad (2)$$

Coverage is marginal over the target strata and prediction randomness.

Shared channel. For Sections 4–5, $a_g = a$ and $b_g = b$ for every g . Known bounds $A, B \geq 0$ satisfy $A + B < 1$, and $(a, b) \in [0, A] \times [0, B]$. Set $\kappa = 1 - A - B > 0$ and assume the observable overlap condition

$$A \leq q_g \leq 1 - B \quad \text{for every } g. \quad (3)$$

Then $p_g = (q_g - a)/(1 - a - b)$ is a valid probability for every channel in the rectangle. Denote the resulting coverage by $F_{ab}(q, r)$, and define

$$H(q, r) = \min_{0 \leq a \leq A, 0 \leq b \leq B} F_{ab}(q, r), \quad V(q, c) = \min_{\substack{r \in [0, 1]^{2m} \\ H(q, r) \geq c}} C(r). \quad (4)$$

This is a population benchmark with known q ; the calibration rule estimates q while retaining known π, A, B .

Heterogeneous channels. Section 6 uses two equal masses, $q = (1 - t, t)$ and $0 < t < 1/2$. It replaces the shared-channel condition with

$$0 \leq a_g, b_g \leq t, \quad |a_1 - a_2| \leq h, \quad |b_1 - b_2| \leq h, \quad 0 \leq h \leq t. \quad (5)$$

Let $V_h(c)$ denote the corresponding robust minimum size. The endpoints $h = 0$ and $h = t$ give shared and unrestricted stratumwise channels, respectively. Label 1 is called favored in stratum 1 and label 0 favored in stratum 2, according to the noisy probabilities. The adjective does not assert that every compatible clean distribution favors those labels.

For each admissible channel, drawing G , then Y from the inverse posterior, and finally Z through the channel constructs a joint law with the same observable (G, Z) distribution. Lower bounds concern algorithms observing this noisy experiment, including exact q , with no additional clean-label side information. Their uniform validity premise averages over calibration as well as testing.

4 OPTIMAL SETS UNDER A SHARED CHANNEL

A common annotation channel is a global constraint. The adverse rate pair must be chosen once for all strata, so minimizing each stratum's contribution separately can give an unnecessarily large set. Nevertheless, the continuum of common rate pairs reduces exactly to four constraints.

Theorem 1 (Shared-channel oracle). *Under the fixed-stratum assumptions of Section 3, the shared channel bounds $A, B \geq 0$, $A + B < 1$, and overlap condition (3), for every $r \in [0, 1]^{2m}$,*

$$H(q, r) = \min_{(a, b) \in \{0, A\} \times \{0, B\}} F_{ab}(q, r).$$

For every $c \in [0, 1]$, $V(q, c)$ is attained and equals the minimum of $C(r)$ subject to the four corner coverage inequalities $F_{ab}(q, r) \geq c$ and the coordinate box constraints. Every randomized stratum-only set has such an inclusion vector, and every inclusion vector is implementable. Any noisy-data algorithm uniformly marginally valid at level c over the compatible shared-channel family has expected size at least $V(q, c)$, even when supplied exact q .

The complete proof is in Appendix A.

The key reduction preserves the common denominator. Put $u_g = r_{g1} - r_{g0}$, $U = \sum_g \pi_g r_{g0}$, $D = \sum_g \pi_g u_g$ and $Q = \sum_g \pi_g q_g u_g$. Then $F_{ab} = U + (Q - aD)/(1 - a - b)$ and

$$\frac{\partial F_{ab}}{\partial a} = \frac{Q - D(1 - b)}{(1 - a - b)^2}, \quad \frac{\partial F_{ab}}{\partial b} = \frac{Q - aD}{(1 - a - b)^2}. \quad (6)$$

Each derivative has a sign independent of the coordinate being varied. First move a to an endpoint without increasing coverage, then move b to an endpoint. This argument minimizes the full weighted sum and avoids an unjustified interchange of minimization and summation.

The remaining proof has two roles. Compactness of the inclusion box and feasibility of the full set establish existence of an optimizer. Equality of the observable data laws then converts any uniformly valid noisy-data procedure into one channel-independent averaged inclusion vector. Its expected size cannot beat the finite program. This final step makes the optimum an information bound within the specified observable experiment, rather than merely a bound on one numerical method.

The program has $2m$ inclusion variables and four coverage inequalities regardless of m . Its generic inclusion formulation is closely related to credal prediction (Javanmardi et al., 2025); the reduction in (6) identifies the constraints for the uncertain forward channel. When $A = B = 0$, it becomes the clean posterior cardinality problem (Sadinle et al., 2016). The overlap condition is essential to treating all four corners as possible worlds. If a noisy stratum lies outside $[A, 1 - B]$, some rate pairs cease to be feasible, and the ambiguity region must be reconsidered.

5 FINITE-SAMPLE CALIBRATION

Estimating the observable probabilities introduces a second source of uncertainty. A direct plug-in solution may sit on the coverage boundary and become invalid after a small estimation error. The size of the necessary correction depends on the policy: including both labels with the same probability makes coverage independent of the posterior.

For each stratum, observe a fixed number $n_g \geq 1$ of independent Bernoulli(q_g) noisy labels. Independence is required within strata; the test observation and prediction randomness are independent of calibration. Let $\alpha, \delta \in (0, 1)$ and define

$$\hat{q}_g = \text{proj}_{[A, 1-B]} \bar{Z}_g, \quad \epsilon = \max_g \sqrt{\frac{\log(2m/\delta)}{2n_g}}, \quad \eta = \frac{\epsilon}{\kappa}, \quad d(r) = \sum_g \pi_g |r_{g1} - r_{g0}|. \quad (7)$$

The policy spread $d(r) \in [0, 1]$ quantifies sensitivity to the clean posterior. Use the calibrated program

$$\hat{r} \in \underset{r \in [0, 1]^{2m}}{\text{argmin}} C(r) \quad \text{subject to} \quad F_{ab}(\hat{q}, r) - \eta d(r) \geq 1 - \alpha \quad \text{at all four corners.} \quad (8)$$

It is a linear program after introducing $z_g \in [0, 1]$ with $z_g \geq r_{g1} - r_{g0}$ and $z_g \geq r_{g0} - r_{g1}$ and replacing $d(r)$ by $\sum_g \pi_g z_g$.

Theorem 2 (Finite-sample coverage and size). *Assume the shared-channel and fixed-stratum conditions of Theorem 1, together with the sampling conditions above. Program (8) is always feasible and admits a measurable optimizer selection. With probability at least $1 - \delta$ over calibration, simultaneously*

$$H(q, \hat{r}) \geq 1 - \alpha, \quad C(\hat{r}) \leq V(q, 1 - \alpha) + [2 - V(q, 1 - \alpha)] \frac{2\eta}{\alpha + 2\eta}.$$

For every admissible true channel, unconditional clean-label coverage is at least $(1 - \delta)(1 - \alpha)$.

The complete proof is in Appendix B.

A simultaneous confidence event controls all fitted policies, without counting the policies that the optimization might select. On $\max_g |\hat{q}_g - q_g| \leq \epsilon$, the exact perturbation identity gives

$$|F_{ab}(\hat{q}, r) - F_{ab}(q, r)| = \left| \frac{\sum_g \pi_g (\hat{q}_g - q_g)(r_{g1} - r_{g0})}{1 - a - b} \right| \leq \eta d(r). \quad (9)$$

A within-stratum exponential bound and a union bound yield event probability at least $1 - \delta$. Projection cannot enlarge the coordinate errors because the true means belong to the projection interval. Equation (9) transfers the penalized estimated constraints to the true corners and hence to every common channel.

For the size comparison, let r^* achieve $V(q, 1 - \alpha)$. Mix it with full inclusion by $r^\lambda = (1 - \lambda)r^* + \lambda \mathbf{1}$, where $\lambda = 2\eta/(\alpha + 2\eta)$. This adds coverage slack and reduces the spread exactly: $d(r^\lambda) = (1 - \lambda)d(r^*)$. The identity $\lambda\alpha = 2\eta(1 - \lambda)$ makes the mixture feasible for the calibrated program on the same event. Optimality then gives the displayed excess-size bound. The full set has spread zero, explaining why feasibility survives even when η is large.

The guarantee is PAC-style: the outer probability concerns calibration, and the inner coverage averages over a fresh test draw. It is distinct from exact marginal conformal coverage at $1 - \alpha$. Uncertain-channel confidence corrections have precedents (Sesia et al., 2023; Penso et al., 2025); the bound above identifies the dependence on κ , sample allocation and inclusion spread for this finite oracle. Small κ amplifies uncertainty, and the least-sampled stratum controls the common radius. The theorem therefore describes convergence toward the population optimum while allowing substantial finite-sample conservatism.

6 EXACT COST OF CHANNEL HETEROGENEITY

A bounded discrepancy between annotation channels permits a quantitative comparison between shared and separate uncertainty. Consider the two-stratum model (5). Symmetry allows a rule with one favored-label inclusion probability x and one other-label inclusion probability y . Its expected size is $x + y$. The challenge is to determine the smallest average favored posterior compatible with the coupled channel constraints, and to show that asymmetric or data-dependent rules cannot improve the resulting bound.

Theorem 3 (Complete coverage–size frontier). *Assume two fixed equally probable strata, $q = (1 - t, t)$, $0 < t < 1/2$, channel rates and differences satisfying (5), and randomized stratum-only prediction with optional empty sets and no additional clean-label side information. For every $c \in [0, 1]$, if $h \leq 1 - 2t$, set $s = 1 - t/(1 - h) \in [1/2, 1]$. Then*

$$V_h(c) = \begin{cases} c/s, & 0 \leq c \leq s, \\ 1 + (c - s)/(1 - s), & s \leq c \leq 1. \end{cases} \quad (10)$$

If $h \geq 1 - 2t$, then $V_h(c) = 2c$. The formulas agree at equality, all optima are attained, and the same lower bounds apply to every noisy-data algorithm uniformly marginally valid over this model, even with exact q .

Appendix E proves the complete result.

Below the threshold, the least-favorable average probability of the favored label is s . Averaging a feasible rule with its group-and-label exchange preserves both size and uniform coverage, giving the symmetric representation. For $x \geq y$, worst coverage is $sx + (1 - s)y$. Writing $z = x + y$ yields the two decisive constraints

$$c \leq sz, \quad c \leq (1 - s)z + 2s - 1, \quad (11)$$

because $x \leq 1$ and $2s - 1 \geq 0$. For $c \leq s$, choose $(x, y) = (c/s, 0)$; for $c \geq s$, choose $(x, y) = (1, (c - s)/(1 - s))$. The branch $x < y$ requires size at least $2c$ and cannot improve either construction. Thus randomization spends the first unit of cardinality on the favored label, then adds the other label once that first inclusion saturates.

At and above the threshold, set $k = 1 - 2t$. The rates $(a_1, b_1, a_2, b_2) = (k, 0, 0, k)$ are admissible because $k \leq h \leq t$, and

$$p_1 = \frac{1 - t - k}{1 - k} = \frac{1}{2}, \quad p_2 = \frac{t}{1 - k} = \frac{1}{2}, \quad F_p(r) = \frac{C(r)}{2}. \quad (12)$$

This single world forces the lower bound $2c$ for every inclusion vector. Including each label with probability c , independently of stratum, attains it for every clean posterior. The observed strata still have different noisy probabilities; the threshold concerns their usefulness for *uniformly valid clean-label prediction*.

Shared versus rectangular channels. The special target $c = 1 - t$ isolates a simple cardinality gap.

Theorem 4 (Cost of separate stratum channels). *Under two fixed equal strata, $q = (1 - t, t)$, and $0 < t \leq 1/3$, the shared-channel optimum with $A = B = t$ at coverage $1 - t$ equals 1. Allowing separate rates in $[0, t]^2$ in each stratum gives optimum $1 + t$. Both optima range over all randomized stratum-only sets. Every noisy-data procedure uniformly marginally valid over the separate-channel family has expected size at least $1 + t$ on this observable distribution, regardless of noisy sample size.*

The complete proof, including its observation-law argument, is in Appendix C.

A shared channel gives the two favored singleton labels coverage $1/2 + (1 - 2t)/(2(1 - a - b)) \geq 1 - t$. For separate channels, each favored posterior ranges from $s_t = (1 - 2t)/(1 - t)$ to one, and

$$\min F_p(x, y) = \begin{cases} s_t x + (1 - s_t)y, & x \geq y, \\ x, & x < y. \end{cases} \quad (13)$$

The choice $(x, y) = (1, t)$ reaches coverage $1 - t$ with size $1 + t$; both branches in (13) give the matching lower bound. This difference is the exact cost of dropping the coupling, rather than a comparison between two heuristics.

Theorem 5 (Linear cost of bounded heterogeneity). *Under two fixed equal strata, $q = (1 - t, t)$, $0 < t \leq 1/3$, and (5), the minimum robust expected size at coverage $c = 1 - t$ is $1 + h$. It is attained by favored-label inclusion one and other-label inclusion h . The lower bound also holds for every uniformly marginally valid procedure observing only the common noisy experiment, even with exact q .*

See Appendix D for the complete proof.

The error probabilities of the favored labels obey

$$e_1 = \frac{t - b_1}{1 - a_1 - b_1}, \quad e_2 = \frac{t - a_2}{1 - a_2 - b_2}, \quad e_1 + e_2 \leq \frac{2t}{1 - h}. \quad (14)$$

Equality is attained at rates $(h, 0, 0, h)$. Thus the favored-posterior endpoint is $s = 1 - t/(1 - h)$, and inserting $c = 1 - t$ into the second branch of (10) gives $1 + h$. The condition $t \leq 1/3$ ensures that all feasible $h \leq t$ remain in the informative regime. These results sharpen the direction of robust-set monotonicity (Javanmardi et al., 2025) into an exact channel-tolerance dependence for this symmetric experiment.

7 NUMERICAL VALIDATION

The numerical study separates population geometry from calibration error. All observations are synthetic Bernoulli strata. The population experiment constructs constraints from admissible forward channels; the calibration experiment resamples noisy labels and evaluates the resulting policies using their exact clean coverage. Thus Monte Carlo variation concerns calibration samples, rather than an additional finite test sample.

Population reconstruction and stress tests. We solve robust linear programs on finite channel grids for $t \in \{0.1, 0.2, 1/3, 0.4, 0.49\}$, 21 values of h between zero and t , and coverage targets drawn from $\{0, 0.2, 0.5, 0.8, 1 - t, 1\}$. Repeated targets are removed, giving 621 programs. Constraints use posterior probabilities obtained by channel inversion; the analytic frontier supplies the comparison value. The maximum absolute cost discrepancy is 2.45×10^{-15} . Twenty seeded random channel-stress batches per (t, h) pair additionally test the predicted policy at target 0.8, with minimum coverage 0.8 to floating-point precision. These finite checks can reveal a violated prediction but cannot certify a continuum of channels.

Figure 1 displays the shared endpoint, the rectangular endpoint and the transition to stratum-independent cost. At $t = 0.2$ and coverage 0.8, the linear specialization gives $1 + h$. At $t = 0.4$, the informative frontier instead reaches 1.6 when $h = 0.2$. This change distinguishes a globally linear dependence at the special target in the lower-noise regime from the

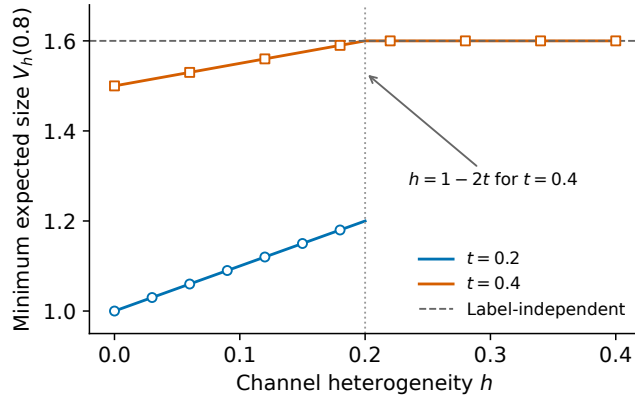


Figure 1: Structural cost at target coverage $c = 0.8$ for two equally weighted synthetic strata with noisy-label probabilities $(1 - t, t)$. Lines show the analytical frontier and open markers show independently solved finite-channel-grid linear programs; the dashed horizontal line is label-independent inclusion with cost $2c = 1.6$. The common-channel endpoint is $h = 0$ and the rectangular endpoint is $h = t$. For $t = 0.2$, the cost is $1 + h$; for $t = 0.4$, it reaches 1.6 at $h = 0.2$. These are deterministic population comparisons; 20 seeded channel-stress batches supplement the grid tests.

complete frontier at stronger noise. Separate shared-channel probes use $m \in \{1, 2, 5, 10\}$ and 80 arbitrary policies; sampled interior channel minima exceed the corner minima, with smallest observed margin 4.05×10^{-6} . The probes supplement the exact four-corner derivation.

Finite calibration. We fix $A = B = 0.2$, $\alpha = 0.2$, $\delta = 0.05$, equal stratum masses, and either $q = (0.8, 0.2)$ or $q = (0.7, 0.3)$. Each stratum receives $n \in \{25, 100, 400, 1600, 6400, 25600\}$ noisy labels. For each of the 12 settings, 20 seeds generate 50 independent binomial replicates each. The corrected program and an uncorrected plug-in program use the same projected estimates. Population oracle size and the theoretical upper bound are reference curves.

Across 12,000 replicates, no corrected policy has robust coverage below 0.8 beyond numerical tolerance. The simultaneous confidence event holds in 11,978 replicates, with no coverage or size-bound violation on that event. For zero failures among 1,000 replicates in a setting, the pointwise Wilson 95% interval is $[0, 0.003827]$. Size intervals use the 20 seed means and a t_{19} multiplier of 2.093; seed reuse across sample sizes makes cross-setting comparisons correlated.

Figure 2 shows the interior case. Uncorrected plug-in failures range from 598 to 646 per 1,000 replicates across the sample sizes. A failure counts any coverage shortfall above tolerance; persistent failure frequency does not imply a nonvanishing shortfall magnitude. At $n = 25600$, corrected mean size is 1.36473 compared with oracle $4/3$. In the boundary case $q = (0.8, 0.2)$, plug-in failures are zero and corrected mean size at the largest n is 1.07576 compared with oracle one. The differing behavior illustrates why the correction’s practical cost depends on the location of the observable probabilities, even with the same rate bounds.

These results support the predicted geometry and reveal conservative finite-sample behavior in the tested regimes. Finite grids include adverse endpoint candidates, so reconstruction is an implementation check rather than an independent proof. The calibration study varies sample size and two observable regimes at one channel rectangle and one pair (α, δ) . It supplies no measurement of annotation channels in natural text or of a trained text classifier.

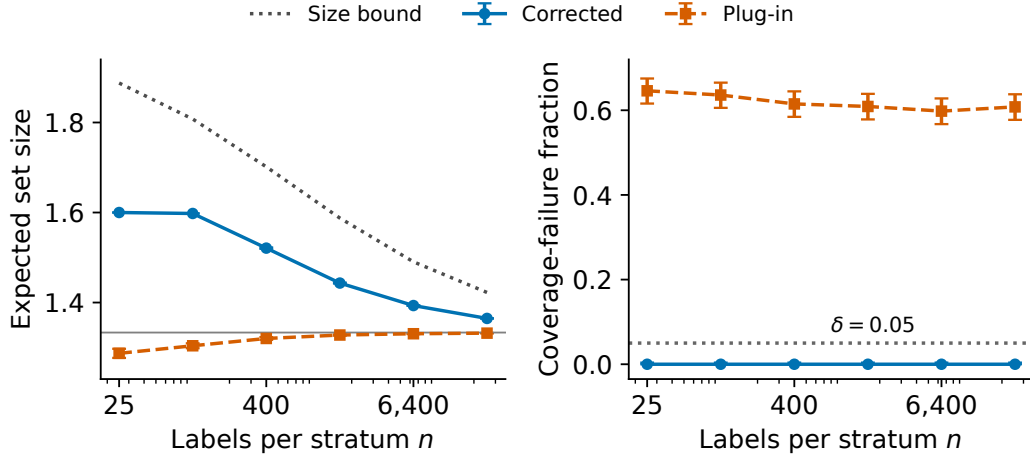


Figure 2: Calibration size and coverage failures for two equal synthetic strata with $q = (0.7, 0.3)$, $A = B = 0.2$, $1 - \alpha = 0.8$ and $\delta = 0.05$. Each sample size uses 20 seeds and 50 binomial calibration replicates per seed. Corrected and plug-in rules share each calibration sample. Left: mean expected set size with pointwise 95% intervals across seed means; dotted curve: analytical size bound; horizontal line: population oracle $4/3$. Right: fraction whose exact worst-channel coverage is below 0.8, with Wilson 95% intervals. Correction removes observed failures at a decreasing size cost. Failure frequency records shortfall occurrence, not magnitude.

8 LIMITATIONS

The fixed stratum map and known target masses are substantive conditions. Learning the map from the same calibration labels or estimating π introduces additional dependence or uncertainty. The shared-channel oracle also assumes every noisy probability lies in $[A, 1 - B]$; outside that overlap region, the feasible channel family changes with q . Finite-sample calibration is proved for a shared channel, whereas the heterogeneous results are population frontiers.

The exact heterogeneity formula uses binary labels, equal masses and mirror-image noisy probabilities. Arbitrary text-dependent annotation errors need not satisfy the difference bounds. The parameter h therefore requires an annotation-based interpretation independent of the noisy probabilities alone. The experiments use synthetic observations and do not assess whether a natural text corpus satisfies these constraints.

Expected size permits randomized empty outputs, and validity is marginal over target strata. Applications requiring nonempty sets or separate guarantees for every subgroup impose different feasible sets. Moreover, a latent clean label presumes a specified target concept; disagreement among legitimate annotators may represent ambiguity in that concept rather than the binary forward corruption process.

9 CONCLUSION

Uncertainty about annotation channels has an efficiency cost that can persist after noisy probabilities are known exactly. In the symmetric binary experiment, bounded channel variation gives an explicit coverage–size frontier, with a fair-label world determining the threshold at which strata cease to improve robust cardinality. A shared-channel four-corner program and a finite-sample confidence correction distinguish this structural cost from calibration error. The natural extension is to characterize least-favorable jointly compatible posteriors for unequal stratum masses and nonsymmetric noisy probabilities, where the group-and-label symmetry no longer reduces the problem to two inclusion variables.

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APPENDIX

A PROOF OF THE SHARED-CHANNEL ORACLE

Proof. We prove Theorem 1. The argument first identifies the decision variables and compatible distributions, then reduces the robust constraint to channel corners, and finally compares arbitrary noisy-data procedures with the population optimum. All probabilities below refer to the fixed target stratum law unless a calibration expectation is explicitly indicated.

Inclusion representation. For a randomized set at stratum g , let I_{gy} be the indicator that label y is included. Conditional independence of test randomization and the fresh label gives

$$\mathbb{E}(|S| \mid G = g) = \mathbb{E}I_{g0} + \mathbb{E}I_{g1} = r_{g0} + r_{g1},$$

and

$$\Pr(Y \in S \mid G = g) = \Pr(Y = 0 \mid G = g)r_{g0} + \Pr(Y = 1 \mid G = g)r_{g1}.$$

Multiplying by the known masses and summing proves (2). No independence between I_{g0} and I_{g1} is used: a set's cardinality is a sum of indicators, and only one of the clean-label cases occurs.

Conversely, given any coordinate vector in $[0, 1]^{2m}$, draw independent uniform variables U_0, U_1 at prediction time and include y precisely when $U_y \leq r_{gy}$. This realizes the given marginals, including coordinates zero or one. The construction may return the empty set; that possibility is permitted by the theorem. Thus the coordinate box is exactly the attainable decision space for the two linear criteria.

Feasibility of each channel world. The forward equation in (1) has the unique solution

$$p_g(a, b) = \frac{q_g - a}{1 - a - b}.$$

The denominator obeys $1 - a - b \geq 1 - A - B = \kappa > 0$. The observable overlap condition gives $q_g - a \geq A - a \geq 0$ and $q_g - a \leq 1 - B - a \leq 1 - b - a$. Therefore $0 \leq p_g(a, b) \leq 1$ at every rate pair in the whole rectangle.

For any such pair, define a distribution by

$$G \sim \pi, \quad Y \mid G = g \sim \text{Bernoulli}(p_g(a, b)),$$

followed by Z through the binary channel with false-positive rate a and false-negative rate b . Its noisy-label mean in each stratum is exactly $a + (1 - a - b)p_g(a, b) = q_g$. Thus every rate pair, including each corner, describes a compatible clean world with the prescribed observable experiment.

A corner attains the minimum. Fix r , and use U, D, Q as defined before (6). Substitution gives

$$F_{ab}(q, r) = \sum_g \pi_g r_{g0} + \sum_g \pi_g (r_{g1} - r_{g0}) \frac{q_g - a}{1 - a - b} = U + \frac{Q - aD}{1 - a - b}.$$

For fixed b , quotient differentiation yields

$$\frac{\partial}{\partial a} \frac{Q - aD}{1 - a - b} = \frac{-D(1 - a - b) + (Q - aD)}{(1 - a - b)^2} = \frac{Q - D(1 - b)}{(1 - a - b)^2}.$$

Its numerator does not depend on a , and its denominator is positive. Consequently F_{ab} is increasing, decreasing or constant throughout $a \in [0, A]$. In all cases there exists $a' \in \{0, A\}$ with $F_{a'b} \leq F_{ab}$.

With this endpoint fixed, differentiation in b gives

$$\frac{\partial F_{a'b}}{\partial b} = \frac{Q - a'D}{(1 - a' - b)^2}.$$

Again the sign is constant along the coordinate interval, so some $b' \in \{0, B\}$ satisfies $F_{a'b'} \leq F_{a'b}$. Each point of the rectangle is therefore dominated in coverage by a corner. Since corners also belong to the rectangle, minimizing over them gives precisely $H(q, r)$. A zero numerator allows either endpoint; $A = 0$ or $B = 0$ merely repeats a corner.

Linear program and existence. At a fixed corner, each $p_g(a, b)$ is a fixed coefficient, making $F_{ab}(q, r) \geq c$ affine in r . The preceding argument proves equivalence between the four inequalities and uniform coverage over the rectangle. Full inclusion $r = \mathbf{1}$ has coverage one under every channel, so the feasible set is nonempty for every $c \in [0, 1]$.

The feasible set is closed and lies in the bounded finite-dimensional coordinate box. Choose a minimizing sequence. A convergent subsequence exists, its limit remains in the box and satisfies all four inequalities by continuity, and its objective equals the infimum because C is continuous. Hence the linear program attains $V(q, c)$. The inclusion construction implements its optimizer.

Noisy-data lower bound. Let $\mathcal{D}$ denote the noisy observations available to any procedure in the stated experiment, and let $R_{gy}(\mathcal{D})$ be its conditional probability of including label y at stratum g , after averaging any additional algorithmic randomness. All compatible worlds induce the same law of $\mathcal{D}$, since their noisy probabilities q and sampling design agree. Define

$$\bar{r}_{gy} = \mathbb{E}_{\mathcal{D}} R_{gy}(\mathcal{D}).$$

This vector is common to every channel. Independence of the fresh test draw gives

$$\mathbb{E}_{\mathcal{D}} C(R(\mathcal{D})) = C(\bar{r}), \quad \mathbb{E}_{\mathcal{D}} F_{ab}(q, R(\mathcal{D})) = F_{ab}(q, \bar{r}).$$

If the procedure has marginal coverage at least c in every world, $\bar{r}$ satisfies every robust coverage constraint. Its cost, which equals the procedure's expected size, is therefore at least $V(q, c)$. Supplying exact q does not change the equality of the observable laws. The bound thus applies even with that information, without asserting finite-sample attainment of the population optimum.

Finally, $c = 0$ allows the zero vector and $c = 1$ allows full inclusion. With $A = B = 0$, the single distinct corner gives the clean-posterior problem. All inverse denominators remain positive under the stated assumptions. These constructions, the corner equality and the observation-law bound establish every assertion of Theorem 1. $\square$

B PROOF OF THE CALIBRATION GUARANTEE

Proof. We prove Theorem 2, using the exact shared-channel representation in Theorem 1. The same calibration event will control validity and cardinality. It is important that the perturbation inequality holds for every inclusion vector simultaneously, because the optimizer is selected from the observed labels.

A simultaneous event with explicit constants. For $Z \sim \text{Bernoulli}(q)$, define $\ell(v) = \log \mathbb{E} \exp\{v(Z - q)\}$ for $v \in \mathbb{R}$. If $q = 0$ or $q = 1$, the centered variable vanishes and $\ell(v) = 0$. Otherwise differentiation of the two-term moment generating function gives

$$\ell'(v) = q_v - q, \quad \ell''(v) = q_v(1 - q_v) \leq \frac{1}{4}, \quad q_v = \frac{qe^v}{1 - q + qe^v}.$$

Since $\ell(0) = \ell'(0) = 0$, the integral Taylor identity, valid for either sign of v , gives

$$\ell(v) = v^2 \int_0^1 (1 - u) \ell''(uv) du \leq \frac{v^2}{8}.$$

For the n_g independent labels within stratum g , moment generating functions multiply. Thus the centered sum $T_g = \sum_{i=1}^{n_g} (Z_{gi} - q_g)$ satisfies $\log \mathbb{E} e^{vT_g} \leq n_g v^2 / 8$. For $e > 0$ and $v > 0$, the elementary bound $\mathbb{E} W \geq z \Pr(W \geq z)$ for nonnegative W gives

$$\Pr(T_g \geq n_g e) \leq \exp(-v n_g e + n_g v^2 / 8).$$

The exponent is minimized at $v = 4e$, yielding $\exp(-2n_g e^2)$. Applying the same argument to $-T_g$ bounds the lower tail identically. Consequently, with

$$e_g = \sqrt{\frac{\log(2m/\delta)}{2n_g}}, \quad \Pr(|\bar{Z}_g - q_g| > e_g) \leq 2e^{-2n_g e_g^2} = \frac{\delta}{m}.$$

A union bound defines the event

$$\mathcal{E} = \bigcap_{g=1}^m \{|\bar{Z}_g - q_g| \leq e_g\}, \quad \Pr(\mathcal{E}) \geq 1 - \delta.$$

This step requires no independence between different strata.

Projection onto $[A, 1 - B]$ cannot increase distance to q_g in that interval. If the empirical mean lies below A , the projected point A lies between it and q_g ; the case above $1 - B$ is identical with the order reversed, and inside the interval nothing changes. Thus on $\mathcal{E}$, $|\hat{q}_g - q_g| \leq e_g \leq \epsilon$ for every g . The event's law depends only on q , hence is the same across compatible channels.

Uniform policy-sensitive transfer. For every channel and every inclusion vector, inverse-channel substitution gives the exact identity

$$F_{ab}(\hat{q}, r) - F_{ab}(q, r) = \frac{\sum_g \pi_g (\hat{q}_g - q_g)(r_{g1} - r_{g0})}{1 - a - b}.$$

On $\mathcal{E}$, triangle inequality and $1 - a - b \geq \kappa$ bound its absolute value by $\epsilon \sum_g \pi_g |r_{g1} - r_{g0}| / \kappa = \eta d(r)$. The only random quantity in this derivation is the coordinate-error bound. Once that bound holds, the inequality applies to the entire inclusion box and every admissible channel; no union over fitted policies is necessary.

Exact linearization, feasibility and measurability. Introduce $z_g \in [0, 1]$ with $z_g \geq r_{g1} - r_{g0}$ and $z_g \geq r_{g0} - r_{g1}$. These imply $z_g \geq |r_{g1} - r_{g0}|$. Since $\eta \geq 0$,

$$F_{ab}(\hat{q}, r) - \eta \sum_g \pi_g z_g \leq F_{ab}(\hat{q}, r) - \eta d(r).$$

Therefore feasibility in the linearized problem implies the original penalized constraints. Conversely, setting $z_g = |r_{g1} - r_{g0}|$ makes the two formulations equal. This proves exact equivalence of their feasible inclusion vectors and objectives.

At full inclusion, $d(\mathbf{1}) = 0$ and $F_{ab}(\hat{q}, \mathbf{1}) = 1$; take $z_g = 0$. Hence the program is feasible regardless of the radius. Its extended feasible set is closed in a compact box, so an optimizer exists by the minimizing-sequence argument in Appendix A. The vector of calibration counts has only $\prod_g (n_g + 1)$ possible values. Choosing one optimizer once for each count vector gives a measurable selection. Ties therefore cause no additional regularity requirement.

On $\mathcal{E}$, any optimizer of the penalized problem satisfies, at every corner,

$$F_{ab}(q, \hat{r}) \geq F_{ab}(\hat{q}, \hat{r}) - \eta d(\hat{r}) \geq 1 - \alpha.$$

The corner equality in Theorem 1 extends this bound to $H(q, \hat{r})$.

A feasible oracle comparison. Write $c = 1 - \alpha$, $V = V(q, c)$, and choose a population optimizer r^* . For every corner, the transfer bound on $\mathcal{E}$ implies

$$F_{ab}(\hat{q}, r^*) - \eta d(r^*) \geq F_{ab}(q, r^*) - 2\eta d(r^*) \geq c - 2\eta,$$

because $d(r^*) \leq \sum_g \pi_g = 1$. The two factors of η have distinct origins: one transfers estimated to true coverage, and the other is the explicit penalty.

Let $\lambda = 2\eta/(\alpha + 2\eta)$ and $r^\lambda = (1 - \lambda)r^* + \lambda \mathbf{1}$. Since $\alpha > 0$ and $\eta \geq 0$, $0 \leq \lambda < 1$. Subtracting the two label coordinates cancels the full-inclusion term, so

$$d(r^\lambda) = (1 - \lambda)d(r^*).$$

By affinity of coverage in r , the penalized value at each corner is at least

$$(1 - \lambda)(c - 2\eta) + \lambda = c + \lambda(1 - c) - 2\eta(1 - \lambda) = c,$$

where the final equality uses $1 - c = \alpha$ and $\lambda\alpha = 2\eta(1 - \lambda)$. Thus r^λ is a feasible competitor. Minimization of C gives

$$C(\hat{r}) \leq C(r^\lambda) = (1 - \lambda)V + 2\lambda = V + (2 - V)\frac{2\eta}{\alpha + 2\eta}.$$

This is the asserted size bound on the same event as validity.

Unconditional coverage and boundary checks. Conditional on any calibration outcome in $\mathcal{E}$, the fresh-test coverage at each admissible true channel is at least c . Off $\mathcal{E}$ it is nonnegative. Integrating gives

$$\Pr(Y_{\text{test}} \in S_{\hat{r}}(G_{\text{test}})) \geq c\Pr(\mathcal{E}) \geq (1 - \alpha)(1 - \delta).$$

The conditioning is on calibration, not on a particular test stratum. Sample sizes as small as one may yield a large radius but leave full inclusion feasible. Degenerate Bernoulli means were covered in the exponential bound. The case $\eta = 0$ gives $\lambda = 0$ without singularity; $\kappa = 0$ and $\alpha = 0$ are excluded by the hypotheses. These observations close the coverage, optimization, measurability and cardinality assertions of Theorem 2. $\square$

C PROOF OF THE SHARED-RECTANGULAR SEPARATION

Proof. We prove Theorem 4 directly for $0 < t \leq 1/3$. This proof compares the shared and rectangular families at the same noisy distribution and coverage target. The policy representation and inverse-channel feasibility follow from the explicit calculations in Appendix A, with $m = 2$, $\pi_1 = \pi_2 = 1/2$, and $A = B = t$. In the rectangular model the inverse calculation is applied separately in each stratum.

Shared-channel upper bound. Use only the favored singleton: label 1 in stratum 1 and label 0 in stratum 2. Expected cardinality is one. For a shared channel, its coverage equals

$$\frac{p_1 + 1 - p_2}{2} = \frac{1}{2} + \frac{q_1 - q_2}{2(1 - a - b)} = \frac{1}{2} + \frac{1 - 2t}{2(1 - a - b)}.$$

The numerator $1 - 2t$ is positive, and $1 - a - b \leq 1$. Thus coverage is at least $1/2 + (1 - 2t)/2 = 1 - t$, with equality at $(a, b) = (0, 0)$. This constructs a feasible shared policy of cost one.

At that no-noise corner, the two posterior vectors are $(t, 1 - t)$ and $(1 - t, t)$. Each label probability is at most $1 - t$. For any inclusion vector, positivity of its coordinates therefore gives

$$F_{00}(q, r) = \frac{1}{2}\{tr_{10} + (1 - t)r_{11} + (1 - t)r_{20} + tr_{21}\} \leq (1 - t)C(r).$$

Uniform coverage $1 - t$ implies $C(r) \geq 1$, proving the matching shared-channel lower bound.

Rectangular posterior range. At $q_1 = 1 - t$, write $p = (q_1 - a)/(1 - a - b)$. Its partial derivatives are

$$\frac{\partial p}{\partial a} = \frac{q_1 + b - 1}{(1 - a - b)^2} = \frac{b - t}{(1 - a - b)^2} \leq 0, \quad \frac{\partial p}{\partial b} = \frac{q_1 - a}{(1 - a - b)^2} \geq 0.$$

Thus its minimum is attained at $(a, b) = (t, 0)$ and its maximum at $(0, t)$:

$$s_t = \frac{1 - 2t}{1 - t} \leq p \leq 1.$$

The same interval describes the favored probability $1 - p_2$ in stratum 2 by exchanging labels and the two channel rates. Every intermediate value can also be realized: varying a from t to zero along $b = 0$ gives the interval $[s_t, 1 - t]$, then varying b from zero to t along $a = 0$ gives $[1 - t, 1]$. Separate stratum channels allow the two favored probabilities to vary independently. Because $t \leq 1/3$, $s_t \geq 1/2$, while $t > 0$ gives $s_t < 1$.

Symmetry reduction. For each compatible posterior pair, the coverage constraint is affine in the four inclusion probabilities. The intersection of these halfspaces with the coordinate box is convex. Simultaneously exchanging strata and labels maps this feasible set to itself: the noisy vector $(1 - t, t)$ is preserved after both exchanges, the rate boxes are unchanged, and equal target masses preserve the objective.

For any feasible vector r , average it with the transformed vector. Convexity preserves feasibility and linearity preserves cost. The average has common favored inclusion x and common other-label inclusion y . Thus restricting to such symmetric policies loses no achievable optimal cost. Its cost is $x + y$. Because the rectangular posterior ranges are independent, its worst coverage is

$$f(x, y) = \min_{v \in [s_t, 1]} \{vx + (1 - v)y\}.$$

When $x \geq y$ the expression is increasing in v , so the minimum is $s_t x + (1 - s_t)y$. When $x < y$, the minimum is obtained at $v = 1$ and equals x . Both adverse choices are attainable by channel choices in the preceding paragraph.

Optimization in both ordering branches. Let $c = 1 - t$. If $x < y$, feasibility gives $x \geq c$, and hence

$$x + y \geq 2c = 2 - 2t \geq 1 + t,$$

where the final inequality is exactly $t \leq 1/3$. The weak bound suffices even though $x < y$ makes the first comparison strict.

If $x \geq y$, put $z = x + y$. Since $2s_t - 1 \geq 0$ and $x \leq 1$,

$$f(x, y) = (1 - s_t)z + (2s_t - 1)x \leq (1 - s_t)z + 2s_t - 1.$$

Consequently

$$z \geq 1 + \frac{c - s_t}{1 - s_t} = 1 + \frac{t^2/(1 - t)}{t/(1 - t)} = 1 + t.$$

The divisions are valid because $t > 0$. Choosing $x = 1$ and $y = t$ attains this cost. Its worst coverage is

$$s_t + (1 - s_t)t = \frac{1 - 2t + t^2}{1 - t} = 1 - t.$$

This verifies feasibility and equality, rather than just a lower estimate on the rectangular optimum.

Irreducibility from noisy observations. Every pair of rectangular channels yields valid posteriors by inversion and the same noisy stratum probabilities. Let a noisy-data algorithm have uniform marginal coverage at level c . Averaging its inclusion probabilities over its data and internal randomness gives one vector $\bar{r}$ common to all compatible worlds. Fresh-test independence implies that this vector's cost equals the algorithm's expected cardinality and that its coverage in each world equals the algorithm's marginal coverage in that world. It is therefore feasible for the rectangular optimization, so its expected size is at least $1 + t$. Exact knowledge of q leaves these worlds indistinguishable and cannot reduce the bound.

At $t = 1/3$, $s_t = 1/2$, and the term $2s_t - 1$ vanishes without changing any inequality. The excluded endpoint $t = 0$ has no noise and the limiting gap is zero. The proof concerns exact uniform marginal validity at c ; a guarantee conditional on a high-probability calibration event must be converted to its unconditional level before using that premise. The constructions and lower bounds prove the shared value one, the rectangular value $1 + t$, and their irreducible gap in Theorem 4. $\square$

D PROOF OF THE LINEAR HETEROGENEITY COST

Proof. We prove Theorem 5 for $0 < t \leq 1/3$ and $0 \leq h \leq t$. The heterogeneous model replaces the common channel by four rates in $[0, t]$ satisfying the two difference bounds. We use the inclusion representation of Appendix A, but derive the heterogeneous coverage bound directly rather than applying a common-channel constraint to different rates.

A bound for the sum of favored-label errors. The favored-label errors are

$$e_1 = 1 - p_1 = \frac{t - b_1}{1 - a_1 - b_1}, \quad e_2 = p_2 = \frac{t - a_2}{1 - a_2 - b_2}.$$

Their numerators are nonnegative and actual denominators are at least $1 - 2t > 0$. Set $u = a_2$ and $v = b_1$. The difference bounds imply $a_1 \leq u + h$ and $b_2 \leq v + h$, so both denominators are at least $d_0 = 1 - h - u - v$. Since $h \leq t$ and $u, v \leq t \leq 1/3$, this auxiliary lower bound is nonnegative.

If $d_0 > 0$, dividing nonnegative numerators by the common lower denominator gives

$$e_1 + e_2 \leq \frac{t - v}{d_0} + \frac{t - u}{d_0} = \frac{2t - u - v}{1 - h - u - v}.$$

For $w = u + v$, direct cross multiplication yields

$$\frac{2t}{1 - h} - \frac{2t - w}{1 - h - w} = \frac{w(1 - h - 2t)}{(1 - h)(1 - h - w)} \geq 0.$$

Both denominators are positive, and $1 - h - 2t \geq 1 - 3t \geq 0$. This proves the claimed error-sum bound in this case.

If $d_0 = 0$, then $h + u + v = 1$ while each term is at most $t \leq 1/3$. Thus necessarily $t = h = u = v = 1/3$. Both error numerators vanish: $t - v = t - u = 0$. The original denominators remain positive, so $e_1 = e_2 = 0$ and the error bound holds directly. This case distinction avoids dividing by a vanishing comparison denominator.

Exact adverse endpoints. Let $\bar{v} = (p_1 + 1 - p_2)/2 = 1 - (e_1 + e_2)/2$ denote the average favored-label probability. The bound gives $\bar{v} \geq s_h = 1 - t/(1 - h)$. Choose

$$(a_1, b_1, a_2, b_2) = (h, 0, 0, h).$$

All rates are at most t , both differences equal h , and both favored-label errors equal $t/(1 - h)$. This attains $\bar{v} = s_h$. At the other endpoint, choosing all four rates equal to t gives $p_1 = 1$ and $p_2 = 0$, hence $\bar{v} = 1$. Probabilities cannot exceed one, so these are the exact minimum and maximum of $\bar{v}$. The inequalities $h \leq t \leq 1/3$ ensure $s_h \geq 1/2$.

Reduction of arbitrary feasible policies. The robust feasible set is convex because each channel four-tuple imposes an affine inequality in the inclusion vector. Simultaneous group and label exchange transforms a channel four-tuple to

$$(a_1, b_1, a_2, b_2) \mapsto (b_2, a_2, b_1, a_1).$$

This preserves the rate box and both difference inequalities. It also preserves the noisy experiment after the exchange and preserves cardinality because masses are equal. Averaging any feasible policy with its image therefore gives a feasible policy of identical cost with favored inclusion x and other inclusion y .

For that vector, coverage in a channel world is $\bar{v}x + (1 - \bar{v})y$. If $x \geq y$, minimizing uses the attained endpoint s_h ; if $x < y$, minimizing uses the attained endpoint one. Thus the exact robust constraint is

$$f_h(x, y) = \begin{cases} s_h x + (1 - s_h)y, & x \geq y, \\ x, & x < y, \end{cases} \quad f_h(x, y) \geq c = 1 - t.$$

No independence between adverse group posteriors is assumed: only the extrema of their compatible average are used.

Matching lower and upper costs. In the branch $x < y$, validity requires $x \geq c$ and size is at least $2c$. The domain gives $1 - 2t \geq h$, hence $2c = 2 - 2t \geq 1 + h$.

In the branch $x \geq y$, write $s = s_h$ and $z = x + y$. Using $s \geq 1/2$ and $x \leq 1$,

$$c \leq f_h(x, y) = (1 - s)z + (2s - 1)x \leq (1 - s)z + 2s - 1.$$

It follows that

$$z \geq 1 + \frac{c-s}{1-s}.$$

The identities $1-s = t/(1-h) > 0$ and $c-s = th/(1-h)$ make the right side $1+h$. Finally the policy $x = 1, y = h$ is in the coordinate box and satisfies

$$f_h(1, h) = s + (1-s)h = 1 - \frac{t}{1-h} + \frac{th}{1-h} = 1-t.$$

It has cost $1+h$, completing both inequalities for the population optimum.

Data-dependent procedures and boundaries. For any admissible rate four-tuple, the inversions define valid probabilities: $q_g \in [t, 1-t]$, the rates are at most t , and $1-a_g-b_g \geq 1-2t > 0$. Drawing the group, its clean label and then its noisy annotation constructs a world with the same observable q . For any uniformly marginally valid noisy-data procedure, average its inclusion vector over this common observation law. Independence of the fresh test draw makes its averaged cost and coverage equal to those of the resulting fixed vector. That vector is feasible in the optimization just solved; the procedure's expected size is therefore at least $1+h$, including when exact q is available.

At $h = 0$ the construction gives the favored singleton rule and cost one. At $h = t$ it gives cost $1+t$, agreeing with Theorem 4. At $t = 1/3$, the comparison denominator can vanish only in the case handled above, while all actual inversions remain defined. These checks close the endpoint and observational assertions and prove Theorem 5. $\square$

E PROOF OF THE COMPLETE COVERAGE-SIZE FRONTIER

Proof. We prove Theorem 3 over the full domain $0 < t < 1/2$, $0 \leq h \leq t$, and $c \in [0, 1]$. The proof does not impose a shared channel or the restriction $t \leq 1/3$. It combines an exact adverse-posterior calculation below the threshold, complete optimization over policy orderings, and a fair-label lower bound above the threshold.

Feasible worlds and decision vectors. Given any allowed rates, let

$$p_1 = \frac{1-t-a_1}{1-a_1-b_1}, \quad p_2 = \frac{t-a_2}{1-a_2-b_2}.$$

The denominators are at least $1-2t > 0$. Each noisy probability lies in $[t, 1-t]$, and $a_g, b_g \leq t$, so $q_g - a_g \geq 0$ and $q_g - a_g \leq 1-a_g-b_g$. Both p_g are therefore valid probabilities. Drawing G uniformly, Y from its posterior and Z through its channel realizes each rate four-tuple as a compatible world.

For a randomized prediction set let r_{gy} be its inclusion marginals. Conditioning on G and Y gives the coverage formula (2), and $\bar{C}(r) = (r_{10} + r_{11} + r_{20} + r_{21})/2$. Independent uniform thresholds implement every vector in $[0, 1]^4$. Full inclusion ensures robust feasibility for every $c \leq 1$. The robust feasible set is an intersection of closed affine halfspaces with this compact box, hence is closed and compact. A minimizing sequence has a feasible convergent subsequence and attains its limiting linear cost. These facts establish existence before the explicit optimizer is determined.

Least-favorable average below the threshold. Suppose $h \leq 1-2t$. Write

$$e_1 = \frac{t-b_1}{1-a_1-b_1}, \quad e_2 = \frac{t-a_2}{1-a_2-b_2}, \quad u = a_2, \quad v = b_1.$$

The difference constraints give $a_1 \leq u+h$ and $b_2 \leq v+h$. Both inverse denominators are consequently bounded below by

$$d_0 = 1-h-u-v \geq 1-h-2t \geq 0.$$

If $d_0 > 0$, nonnegativity of the numerators and direct subtraction give

$$e_1 + e_2 \leq \frac{2t-u-v}{1-h-u-v}, \quad \frac{2t}{1-h} - \frac{2t-u-v}{1-h-u-v} = \frac{(u+v)(1-h-2t)}{(1-h)(1-h-u-v)} \geq 0.$$

If $d_0 = 0$, then $u + v = 1 - h \geq 2t$, while $u + v \leq 2t$. Hence $h = 1 - 2t$ and $u = v = t$. Both error numerators are zero, and the positive original denominators imply $e_1 = e_2 = 0$. In either case,

$$\bar{v} := \frac{p_1 + 1 - p_2}{2} = 1 - \frac{e_1 + e_2}{2} \geq s := 1 - \frac{t}{1 - h}.$$

Since $1 - h \geq 2t > 0$, $s \geq 1/2$; since $t > 0$, $s < 1$.

The rates $(h, 0, 0, h)$ satisfy the box and difference bounds, and give both errors $t/(1 - h)$, attaining $\bar{v} = s$. The rates (t, t, t, t) give $p_1 = 1$, $p_2 = 0$, attaining $\bar{v} = 1$. Thus the exact extrema are s and one. The argument requires these extrema only; it does not require a claim that every intermediate average is attained.

Symmetry without loss of optimality. For a rate four-tuple θ , denote its coverage functional by $F_\theta(r)$. The group-and-label involution takes $\theta = (a_1, b_1, a_2, b_2)$ to $T\theta = (b_2, a_2, b_1, a_1)$. Both rate differences and all rate bounds are preserved, as are the noisy probabilities after relabeling. The corresponding transformation of inclusion vectors satisfies $F_\theta(Tr) = F_{T\theta}(r)$ and $C(Tr) = C(r)$.

If r is uniformly feasible, then Tr is uniformly feasible. By convexity, $(r + Tr)/2$ is feasible and has the same cost. It takes the form

$$r_{11} = r_{20} = x, \quad r_{10} = r_{21} = y, \quad 0 \leq x, y \leq 1.$$

For a channel with average favored probability $\bar{v}$, its coverage is $y + (x - y)\bar{v}$ and cost is $x + y$. The attained extrema from the preceding step imply

$$\inf_{\theta} F_{\theta}(r) = \begin{cases} sx + (1 - s)y, & x \geq y, \\ x, & x < y. \end{cases}$$

This equality uses the joint feasible family. It does not minimize two group contributions independently.

Complete optimization when $x \geq y$. Set $z = x + y$. Since $s \geq 1/2$ and $y \geq 0$,

$$sx + (1 - s)y \leq s(x + y) = sz.$$

Also $x \leq 1$ gives

$$sx + (1 - s)y = (1 - s)z + (2s - 1)x \leq (1 - s)z + 2s - 1.$$

A feasible vector must therefore satisfy

$$z \geq \frac{c}{s}, \quad z \geq \frac{c - 2s + 1}{1 - s}.$$

Both frontier denominators are positive: $s > 0$ and $1 - s > 0$. Subtracting the first lower bound from the second gives

$$\frac{c - 2s + 1}{1 - s} - \frac{c}{s} = \frac{(2s - 1)(c - s)}{s(1 - s)}.$$

For $c \leq s$, the first bound is the larger one; for $c \geq s$, the second is the larger one. At $s = 1/2$ they coincide for all c .

For $0 \leq c \leq s$, choosing $x = c/s$ and $y = 0$ gives coordinates in the required range, cost c/s and coverage c . For $s \leq c \leq 1$, choose $x = 1$ and $y = (c - s)/(1 - s)$. Again $0 \leq y \leq x \leq 1$, and cost and coverage are respectively $1 + (c - s)/(1 - s)$ and c . These choices attain the larger lower bounds throughout the two intervals.

Exclusion of an improvement from $x < y$. In this branch the exact worst coverage is x . Validity gives $x \geq c$ and hence $x + y \geq 2c$. For $c \leq s$, the constructed cost obeys $c/s \leq 2c$ because $s \geq 1/2$. For $c \geq s$,

$$2c - \left(1 + \frac{c - s}{1 - s}\right) = \frac{(2s - 1)(1 - c)}{1 - s} \geq 0.$$

Thus the reverse ordering cannot improve either branch. Symmetrization and the exhaustive ordering split prove the stated global optimum below the threshold over all randomized inclusion vectors.

Fair-label world above the threshold. Now suppose $h \geq 1 - 2t$, and set $k = 1 - 2t$. The assumptions give $0 < k \leq h \leq t$, so $\theta = (k, 0, 0, k)$ is admissible. Its posteriors are

$$p_1 = \frac{1 - t - k}{1 - k} = \frac{t}{2t} = \frac{1}{2}, \quad p_2 = \frac{t}{1 - k} = \frac{1}{2}.$$

For an arbitrary, possibly asymmetric inclusion vector,

$$F_\theta(r) = \frac{1}{4}(r_{10} + r_{11} + r_{20} + r_{21}) = \frac{C(r)}{2}.$$

Uniform coverage at level c therefore forces $C(r) \geq 2c$. Conversely, set all four inclusion coordinates equal to c . Then $F_\theta(r) = c$ for every pair of valid clean posteriors, and $C(r) = 2c$. This policy attains the lower bound over the whole admissible family.

At $h = 1 - 2t$, $s = 1/2$. The first informative branch is $c/s = 2c$, and the second is $1 + (c - 1/2)/(1/2) = 2c$. Thus both descriptions agree at equality. If $t < 1/3$, then $1 - 2t > t$, so no admissible h reaches this second regime.

Observation-law lower bound and final endpoints. All compatible worlds constructed at the start have identical noisy observation laws. For any procedure using only those observations and independent randomization, average its inclusion probabilities over that common law. The resulting vector is independent of the chosen world. The linear formulas show that its cost equals the procedure's expected size and its coverage in each world equals that procedure's marginal coverage. Uniform validity places this vector in the robust feasible family already optimized. Therefore the same lower bounds apply to all such noisy-data procedures, even when exact q is supplied.

At $c = 0$, the first construction is empty inclusion with cost zero. At $c = s$, both informative constructions give a favored singleton and cost one. At $c = 1$, full inclusion has cost two. The original inverse denominators remain positive for every $t < 1/2$, and the only zero auxiliary denominator was handled without division. Together, the attainable policies, branchwise lower bounds, threshold construction and observational averaging establish all conclusions of Theorem 3. $\square$